

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8372**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600002
Course Section:	1				

L_61_480_986_L_1505_4837



Name of the candidate. Anna Jogekar

Candidate's UID number:

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Examination class room number:

2	0	2
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Invigilator's signature with date:

 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE - BTECH - CSE
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: CSE - AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wed 8/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

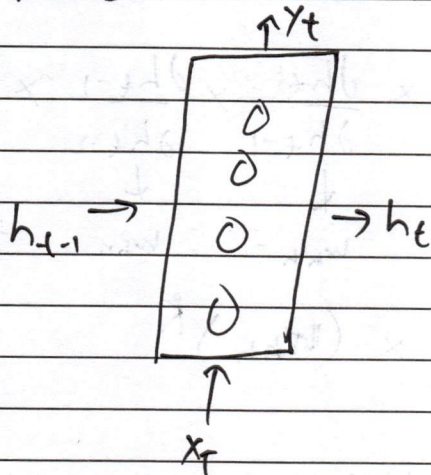
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0-3]

a]

RNN : Recurrent Neural Network



$$h_t = w_{hh} \cdot h_{t-1} + w_{hx} \cdot x_t + b_t$$

$$y_t = w_{yh} \cdot h_t + b_y$$

$$\frac{\partial L}{\partial h_t} = \frac{\partial}{\partial h_t} (w_{yh} \cdot h_t + b_y)$$

$$\frac{\partial L}{\partial h_t} = w_{yh} \rightarrow \textcircled{1}$$

$$\frac{\partial L}{\partial h_{t-1}} = \frac{\partial}{\partial h_{t-1}} (w_{yh} \times (w_{hh} \cdot h_{t-1} + w_{hx} \cdot x_t + b_t))$$

$$\frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial h_t} \times \frac{\partial h_t}{\partial h_{t-1}}$$

$$= w_{yh} \times \frac{\partial}{\partial h_{t-1}} (w_{hh} \cdot h_{t-1} + w_{hx} \cdot x_t + b_t)$$

$$= w_{yh} \times [w_{hh} + w_{hx} \frac{\partial x_t}{\partial h_{t-1}}]$$

$$= w_{yh} \times w_{hh}$$

$$\frac{\partial L}{\partial h_{t+2}} = \frac{\partial L}{\partial h_t} \times \frac{\partial h_{t+1}}{\partial h_t} \times \frac{\partial h_{t-1}}{\partial h_{t+2}}$$

→ Similar to above, we find $\frac{\partial h_{t-1}}{\partial h_{t+2}} = w_{hh}$

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So ~~for~~ moving ahead,

$$\frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial h_t} \times \frac{\partial h_t}{\partial h_{t-1}} \times \frac{\partial h_{t-1}}{\partial h_{t-2}} \times \dots \times \frac{\partial h_{t-k+1}}{\partial h_{t-k}}$$

$$\downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \dots$$

$$w_{yh} \quad w_{hh} \quad w_{hh} \quad \dots$$

$$\frac{\partial L}{\partial h_{t-k}} = w_{yh} \times (w_{hh})^k$$

This is the gradient expression.

If $w_{hh} < 1$ then as k becomes larger, $(w_{hh})^k \rightarrow 0$.

This is called vanishing gradients.

LSTM has a cell memory C_t which can hold information for long time.

The forget gate decides which information to be removed from C_t . This is how LSTM overcomes vanishing gradients.

$$f_t = w_f(x_t, h_{t-1}) + b_f$$

$$C_t = C_{t-1} \times f_t + i_t \cdot \tilde{C}_t$$

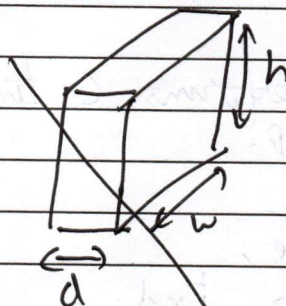
The term $C_{t-1} \times f_t$ decides what info to be kept at every step and it is kept in its entirety. This avoids the vanishing gradient problem.

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Q-2]

a)



Standard 2D convolution:

$$w_{out} = \left\lfloor \frac{w - F + 2P}{S} \right\rfloor + 1$$

Assume $P=0, S=1$

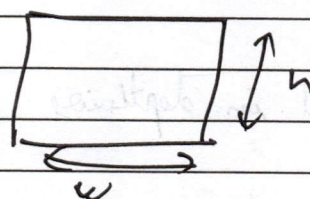
$$w_{out} = w - F + 1$$

$$h_{out} = h - F + 1$$

$$d_{out} = \text{no. of filters} = k$$

Depthwise separable convolution:

Standard 2D convolution:



$$w_{out} = \left\lfloor \frac{w - F + 2P}{S} \right\rfloor + 1$$

Assume $P=0, S=1$

$$w_{out} = w - F + 1$$

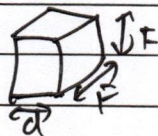
$$h_{out} = h - F + 1$$

For a depth d , This operation is done by one filter d times.

→ For k filters, $k \times d \times w_{out} \times h_{out}$ convolutions performed.

Depthwise separable convolution:

→ In this, we perform 3D convolution by taking depth of filter as d .



$$w_{out} = w - F + 1$$

$$h_{out} = h - F + 1$$

$$d_{out} = k$$

$$\begin{aligned} \text{No. of convolutions} &= w_{out} \times h_{out} \times d_{out} \\ &= w_{out} \times h_{out} \times k \end{aligned}$$

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For input size $H \times W \times C$,

Depthwise separable convolution performs C times less convolutions than standard 2D.

→ Also as no. of layers increase,
in standard → new depth was $k \times d$
whereas in depthwise → d

in every new step, standard will perform exponentially more no. of convolutions.

Hence, depthwise separable convolution is critical in modern architecture.

No. of parameters to be trained in depthwise
→ $F \times F \times k$

No. of parameters in 2D
 $= F \times F \times k \times C$

Hence training depthwise separable convolution is parameter efficient.

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Q-1]

b]

For a given model which predicts $\hat{y}$ on input $x_1, x_2, \dots$ and actual output y ,

$$\text{Mean squared error} = \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$$

$$\hat{y} \text{ is given as } \sum_{i=1}^n w_i x_i + b_i$$

If model is overfitting, i.e. $\hat{y} \approx y$ for all training inputs, this is because there may be too many hidden layers or no. of neurons in hidden layer. The model learns to predict a given value for an input set exactly but does not learn the underlying approximation or function.

~~Bias - variance tradeoff:~~

If model capacity is high, i.e. it can learn very complex patterns and we want it to learn a simple pattern, then it overfits.

→ Solution - perform regularization, or make model less complex.

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Q-3)

b) Gated Recurrent Unit (GRU):

These have 2 gates:

1) Reset gate

$$r_t = \sigma(w_r[x_t, h_{t-1}] + b_r)$$

This gate chooses what to remove from memory.

2) Update gate:

$$z_t = \sigma(w_z[x_t, h_{t-1}] + b_z)$$

→ This gate chooses what to add to memory.

The candidate hidden state is given as:

$$\tilde{z}_t = r_t \cdot h_{t-1} + z_t(x_t)$$

The final hidden state is given as:

$$h_t = (1 - \tilde{z}_t) \cdot h_{t-1} + \tilde{z}_t \cdot \tilde{z}_t$$

GRU's have fewer parameters than LSTM because

GRU only has 1 hidden state whereas

LSTM → has long term (c) and short term (s),
i.e. two hidden states.

→ Real world use:

→ GRU is computationally faster than LSTM
also takes less time to train.

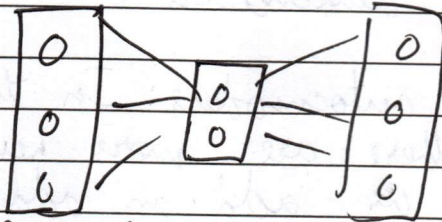
→ Fewer no. of parameters need to be trained

→ Application in text generation.

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Q-2)
b)



Undercomplete Autoencoder:

In this, the no. of neurons in latent space is less ~~but~~ than the input.

Let's take an input function x and decoder function $z = f(x)$

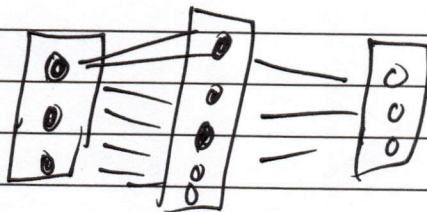
The encoder predicts $\hat{x} = g(z)$ and we use a MSE for loss function.

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n \|x - \hat{x}\|^2$$

→ This is the same loss function when using PCA to find first k ($k < n$) components → corresponding to the latent space.

Over complete autoencoder:

→ In this the no. of neurons in latent space $\geq$ no. of neurons in input.



With extra space available the model does not need to learn important parts of input when it can just copy it directly to its memory.

The encoder learns to copy entire input and decoder copies it from latent space to output. This means it's actually learning the identity function.

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Why regularization is required with overcomplete autoencoders.

→ eg : Sparse autoencoders: In this even though there are more neurons in latent space, we add an additional loss function which measures difference between no. of neurons active for a given input to a low no. of neurons (say k).

Hence, ~~very~~ very few neurons actually fire for any given input and ~~does~~ we limit it from learning the identity function.

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Q-1]

$$a) f(w) = \frac{1}{2} \|x_w - y\|^2$$

$$\frac{\partial f}{\partial w} = \frac{\partial}{\partial w} \left(\frac{1}{2} (x_w - y)^2 \right)$$

$$= \frac{1}{2} \times 2 \|x_w - y\| \times \frac{\partial}{\partial w} (x_w - y)$$

$$= \|x_w - y\| \times \frac{\partial x_w}{\partial w}$$

Gradient update rule is

$$w \rightarrow w - \eta \frac{\partial f}{\partial w} = w - \eta \|x_w - y\| \frac{\partial x_w}{\partial w}$$

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